BOATWORTHY MARINE SERVICES, LLC
MARINE SURVEYOR AND CONSULTANT

1972 Hatteras 42 Convertible

"LONGSHANKS"

MEMBER OF SOCIETY OF ACCREDITED MARINE SURVEYORS (SAMS),
AMERICAN BOAT AND YACHT COUNCIL (ABYC),
NATIONAL MARINE ELECTRONICS ASSN. (NMEA), AND BoatUS.

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Boatworthy
Marine Services, LLC

Report of Marine Survey

Of The Vessel

"LONGSHANKS"

1972 Hatteras 42 Convertible

Conducted by
Ronald E. Varg

INDEPENDENT MARINE SURVEYOR

PREPARED EXCLUSIVELY FOR:
Edward Marney

8/21/2024

MEMBER OF SOCIETY OF ACCREDITED MARINE SURVEYORS (SAMS),
AMERICAN BOAT AND YACHT COUNCIL (ABYC),
NATIONAL MARINE ELECTRONICS ASSN. (NMEA), AND BoatUS.

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I. INTRODUCTION

SCOPE OF SURVEY

Acting at the request of Edward Marney, the attending surveyor did attend onboard the *1972 Hatteras 42 Convertible, "LONGSHANKS"* on August 20, 2024 at 10 am, where an ''in-the-water-survey'' was conducted at Safe Harbor at Beaufort Downtown Marina, Beaufort, SC. The ship's papers were on board and appeared to be in order. The Hull Identification Number was verified from a placard mounted in the engine room. A sea trial or out-of the water inspection of underwater machinery was not performed. The reason for the survey, was to ascertain the physical condition and value of the vessel for insurance purposes. AC and DC power were used to check operation of the electrical systems specified in this report only.

No reference or information in this report should be construed to indicate evaluation of the internal condition of the engines or the propulsion system's operating capacity.

Electronic equipment was checked for ''power up'' only.

This vessel was surveyed without removal of any parts, including fittings, tacked carpet, screwed or nailed boards, anchors and chain, fixed partitions, instruments, clothing, spare parts and miscellaneous materials in the bilges and lockers, or other fixed or semi-fixed items. Locked compartments or otherwise inaccessible areas would also preclude inspection. Owner is advised to open up all such areas for further inspection. Further, no determination of stability characteristics or inherent structural integrity has been made and no opinion is expressed with respect thereto. This survey report represents the condition of the vessel on the above dates, and is the unbiased opinion of the undersigned, but it is not to be considered an inventory or a warranty either specified or implied.

NOTE: It is recommended and understood that all DIESEL engines be surveyed by a qualified Engine Surveyor to determine the internal condition of the engines, gears and pumps, heat exchangers, coolers, etc.

Where installation of double hose clamps are recommended throughout this report, it is understood that double clamps should only be installed where there is sufficient length of tailpiece/pipe available and hose length overlap to allow correct installation. No clamp shall be installed closer than 1/4" to the end of the hose and must fully engage the tailpiece/pipe or fitting. Any clamp extending over the end may cause the hose to be cut internally or force the hose off the fitting and is an incorrect installation.

Solid, unperforated type hose clamps utilizing 316 stainless steel with rolled edges are recommended for all hose to pipe joints particularly when located below the waterline.

I. INTRODUCTION

CONDUCT OF SURVEY :

THE MANDATORY STANDARDS PROMULGATED BY THE UNITED STATES COAST GUARD (USCG), UNDER THE AUTHORITY OF TITLE 46 UNITED STATES CODE (USC); TITLE 33 AND TITLE 46, CODE OF FEDERAL REGULATIONS (CFR), AND THE VOLUNTARY STANDARDS AND RECOMMENDED PRACTICES DEVELOPED BY THE AMERICAN BOAT AND YACHT COUNCIL (ABYC) AND THE NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) HAVE BEEN USED AS GUIDELINES IN THE CONDUCT OF THIS SURVEY BUT, COMPLETE COMPLIANCE WITH SUCH STANDARDS VARIES WITH THE INTENDED SERVICE OF THE VESSEL, AND IS NOT GUARANTEED.

Use of asterisks * in the body of the report will indicate that a finding will be listed in the Findings and Recommendations section pertaining to the asterisked item, following the body of the report. The deficiencies reported herein reflect the conditions observed at the time the survey was conducted.

Images supplied with this report were produced with an Olympus Tough T-6 digital camera and represent a true and accurate representation of the subject at the time the image was taken.

NOTE:
1. This report is issued for the exclusive use of the individual(s), financial institution(s) and/or insurance company (ies) as may be specifically identified (named) upon this surveyor's report and may contain information that is privileged, confidential and exempt from disclosure under applicable law. Any entities or persons that are not identified herein are hereby advised that any dissemination, distribution or copying of this report is strictly prohibited; no such entity or person shall have any right to rely upon the contents of this surveyor's report.

2. In the event that this surveyor is called upon, after rendering a Marine Survey Report, to explain, modify or supplement the report, or its contents, or should the surveyor be called upon to render expert advise, testimony or to provide survey expertise in any dispute in litigation (or not), the surveyor will be compensated by the owner/insured in accordance with the fees customarily charged in the surveying industry.

LIMITED LIABILITY:
Acceptance and use of this report by the client acknowledges the client's understanding that the report has been composed of information that is believed to be true after reasonable investigation and inquiry but is not warranted to be so. The information was obtained without drilling, diving, ultrasonics, cleaning or opening up to expose parts or conditions ordinarily concealed. There were no tests for tightness or soundness conducted other than the conditions noted visually. Acceptance and use of this report acknowledges the client's understanding that no determination of stability or structural strength has been made and no opinion is expressed. Acceptance and use of this report acknowledges the client's understanding that Boatworthy Marine Services, LLC,. does not accept any responsibility for damage or deterioration not found or discovered during the course of survey, nor for consequential damage, deterioration or loss due to any error or omission. The Client hereby undertakes to keep the Surveyor/Consultant and its employees, agents and subcontractors indemnified and to hold them harmless against all actions, proceedings, claims, demands or liabilities whatsoever or howsoever arising which may be brought against them or incurred or suffered by them, and against and in respect of all costs, loss, damages and expenses (including legal costs and expenses on a full indemnity basis) which the Surveyor/Consultant may suffer or incur (either directly or indirectly) in the course of the services under these Conditions. Notwithstanding the above clause, in the event that the Client proves that the loss, damage, delay or expense was caused by the negligence, gross negligence or willful default of the Surveyor/Consultant aforesaid, then, save where loss, damage, delay or expense has resulted from the Surveyor's/Consultant's personal act or omission committed with the intent to cause same or recklessly and with knowledge that such loss, damage, delay or expense would probably result, the Surveyor's/Consultant's liability for each incident or series of incidents giving rise to a claim or claims shall never exceed a sum calculated on the basis of ten times the Surveyor's/Consultant's charges.

ATTENDING SURVEYOR:
Ronald E. Varg

II. GENERAL INFORMATION

GENERAL INFORMATION

FILE NUMBER:	**41VIKING88_23_06**
SURVEY PREPARED FOR:	**Edward Marney**

NAME OF VESSEL:	**"LONGSHANKS"**
TYPE OF SURVEY:	**Insurance**
OVERALL VESSEL RATING: ....****	**ABOVE AVERAGE**
ESTIMATED MARKET VALUE: ....**	**$86,600.**
ESTIMATED REPLACEMENT COST:..............****	**$2,025,000.**
YEAR/MAKE/MODEL OF VESSEL:	**1972 Hatteras 42 Convertible**
BUILDER:	**Hatteras Yachts, New Bern, NC**
YEAR BUILT:	**1972**
HULL IDENTIFICATION NUMBER (HIN):	**Pre HIN numbers**
HULL NUMBER:	**319**
HAILING PORT:	**Newport RI**
USCG DOCUMENTATION NUMBER: ....***	**928034 (Expired)**
OWNER'S NAME: ....***	**SSF Limited LLC**
OWNER'S ADDRESS:	**5600 Post Rd #114-120, East Greenwich, RI**
PLACE OF SURVEY:	**Safe Harbor at Beaufort Downtown Marina, Beaufort, SC**
DATE/TIME OF SURVEY:	**8/20/2024, between 10 am and 1:30 pm**
HULL MATERIAL:	**FRP (Fiber Reinforced Plastic).**
HULL TYPE:	**Modified V with moderate deadrise aft.**
LENGTH OVER ALL (L.O.A).: ....*	**42' 8"**
BEAM: ....*	**13' 10"**
DRAFT: ....*	**3' 5"**
DISPLACEMENT: ....*	**31,000 lbs.**
OVERHEAD CLEARANCE:	**19' 8" (reportedly)**
GROSS/NET TONS: ....***	**27/22 T**
PROPULSION SYSTEM:	**Twin Detroit Diesel 6-71N engines**
FUEL TYPE:	**Diesel.**
FUEL CAPACITY: ....*	**400 gal.**
AC POWER:	**Yes, Two (2) 125 volt, 30 amp. Inlets**
DC POWER:	**12 volt.**
FRESH WATER CAPACITY: ....*	**150 gal.**

GENERAL INFORMATION*(continued)*

HOLDING TANK: . . . .* **125 gal.**

INTENDED CRUISING AREA: . . . . **Near coastal South Carolina and Florida.**

INTENDED USE: . . . . **Recreational Coastal Cruising & Diving**

Asterisks * in this General Information section refers to the source of such information as follows:
* Per Manufacturer's Specifications
**Refer to Summary and Valuation Section
*** Per USCG Documentation
**** Per Buc Book

DEFINITION OF TERMS

The terms and words used in this report have the following meanings as used in this *Report of survey:*

APPEARS:
Indicates that a very close inspection of the particular system, component or item was not possible due to constraints imposed upon the surveyor(e.g. no power available, inability to remove panels, or requirements not to conduct destructive tests).

FIT FOR INTENDED USE:
Use which is intended by Survey Purchaser(present or prospective owner).

SERVICEABLE: ADEQUATE:
Sufficient for a specific requirement.

POWERS UP:
Power was applied only. This does not refer to the operation of any system or component unless specifically indicated.

EXCELLENT CONDITION:
New or like new.

GOOD CONDITION:
Nearly new, with only minor cosmetic or structural discrepancies noted.

FAIR CONDITION:
Denotes that system, component or item is functional as is with minor repairs. (MONITOR OFTEN)

POOR CONDITION:
Unusable as is. Requires repairs or replacement of system, component or item to be considered functional.

SEA TRIAL:
In the context of this survey, refers to operating and testing the vessel in a convenient body of water such as a sound, river, lake, or waterway which would allow sufficient room to safely operate and maneuver under sail, or at Wide Open Throttle (WOT), in conditions not necessarily similar to the open sea.

USE OF *:
Use of * in the body of this report will indicate that a finding will be listed in the "*Findings and Recommendations*" section pertaining to the * item.

HULL DECK AND SUPERSTRUCTURE

HULL CONSTRUCTION

TYPE: Modified-V, planing type, with stepped shear, flared bow, and moderate deadrise.

MATERIAL: Solid handlaid FRP (fiber reinforced plastic).

EXTERIOR HULL: White gelcoat with navy blue boot stripe. White Emron paint.

BULKHEADS: Athwartships reinforcement enhanced by bulkheads bonded to the hull with FRP (fiber reinforced plastic). Appears serviceable where sighted.

STRINGERS: Hull stiffness provided by FRP longitudinal stringers. Complete inspection not possible due to limited access. Appears serviceable where observed.

STEM: Curved stem, no nicks or signs of damage.

TRANSOM: Reinforced FRP, slightly rounded with attached composite swim platform and retractable swim ladder. Serviceable.

BILGE: Deep (below decks) bilge area provides the area for most boat systems and tankage. Generally very clean.

CHAIN LOCKER (DRAINAGE): Drainage to bilge, size adequate, access good, location forward bow.

LIMBER HOLES: Limber holes are of adequate size and clear where sighted.

DECK CONSTRUCTION

TYPE: Molded & cored FRP (fiber reinforced plastic) with white paint and non-skid surface in good condition.

COCKPIT: Molded FRP (fiber reinforced plastic) with white gelcoat and non-skid surface.

HULL-TO-DECK JOINT

TYPE: Visible from the forepeak in the chain locker the hull to deck joint was of the overlap (Coffee can) flange type with screw stainless fasteners on estimated 6" centers through a stainless steel rub rail. Appeared serviceable where sighted.

DECK FITTINGS

BOW PULPIT (BOW RAIL): Welded 1-1/2"aluminum tubing rail system, runs from the bow aft to the superstructure. Serviceable and securely mounted.

TOE RAILS: Molded FRP 4" wide toe rails, part of deck layup. Serviceable.

VENTILATION: Three hatches, one (1) over the galley, one (1) over the master cabin, and one (1) forward above V-berth. All latches and hinges are serviceable. Sliding windows each side of the main cabin in serviceable condition.

BOWSPRIT: FRP attached platform, with stainless roller insert for single anchor. Serviceable.

SCUPPERS: Cockpit has scuppers at port and starboard aft corners approx. 1-1/2" x 6" with drainage through the transom. Serviceable.

CHOCKS AND CLEATS: Eight (8) Stainless steel 10" inch cleats, four on each side. Also 2 chocks, 5" stainless steel at the transom. All sighted were serviceable.

WINDLASS/GIPSY: Lawrence Simpson, two direction electric windlass. Did not test.

DECK SURFACE: White gel coat with molded in non-skid. Condition is serviceable.

GRAB RAIL: Stainless hand rails along sides of super structure and others at various locations on vessel. Appears adequate.

HULL DECK AND SUPERSTRUCTURE

DECK FITTINGS*(continued)*

CLEATS: Chocks and cleats appeared to be stainless steel all sighted were thru-bolted and serviceable.

SUPERSTRUCTURE

MATERIAL: Cabin house and deck are one unit molded FRP (fiber reinforced plastic).

DECK HATCHES: Three (3), 19" X 19" in forward cabin deck area. Serviceable.

WINDOWS/PORTS/DOORS: The sides of cabin house have large stationary & sliding windows. The door to the main salon opens out into the cockpit. Additional cabin light is supplied by an aft window. All serviceable.

Access to flybridge

Cockpit

FITTINGS AND HARDWARE: Various stainless handles mounted in strategic places. A large 8 step ladder allows access to the flybridge from the cockpit. Serviceable.

JOINERY STRESS: None Sighted.

CANVAS AND SUPPORT STRUCTURE: White isinglass curtains, 4 sides on flybridge, with white FRP top in good condition.

SUPERSTRUCTURE HOUSE TO DECK JOINT: Deck house and deck appeared to be molded seamlessly, no joint was observed.

COCKPIT: Spacious cockpit deck has a door to the main cabin, a stepladder leading to the fly bridge, storage forward, and access to each side weather deck.

BRIDGE DECK

MATERIAL: FRP (fiber reinforced plastic) molded on fly bridge, serviceable.

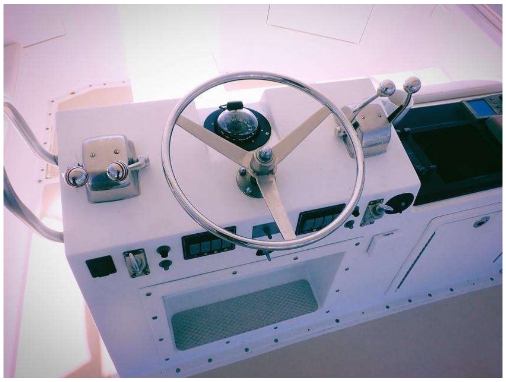
TYPE: The bridge provides a helm station with two (2), white upholstered pedestal mounted helm chairs. All steering, navigation electronics, and engine controls and gauges appear serviceable.

HULL DECK AND SUPERSTRUCTURE

BRIDGE DECK*(continued)*

* **TYPE:** ***(continued)***

Helm seating



Helm station

SEATS: Pedestal mounted swivel chairs are provided for the helmsman and mate. Good condition.

BIMINI: White FRP with welded aluminum tubular support structure in very good condition.

WINDSHIELD: A framed 5 panel clear plastic spray shield. Serviceable.

Captains view fwd

Bow looking aft

SAFETY RAIL SYSTEM: Aluminum tubular rail and stanchion system surrounds the upper helm area. Appears serviceable.

OTHER: A fold down stair system was added to allow access to the foredeck from the flybridge. Serviceable.

ADDITIONAL EQUIPMENT AND ACCESSORIES

ACCESSORIES: Spreader and cockpit lights provide additional and convenient task lighting.

DINGHY/TENDERS: Zodiac, 10' 6" inflatable with inflatable floor, stowed in lazarette and not inspected.

FENDERS: Ample pneumatic cylindrical type fenders. Serviceable

DOCK LINES: Several black braided dock lines sighted. Serviceable.

CABIN APPOINTMENTS

INTERIOR DESCRIPTION:

JOINERY AND FINISH: The joinery and finish of the teak and Maple interior was in good condition..

Main Salon

Master cabin

CABIN BRIGHT WORK: Satin varnish finish on all teak doors and trim. Good condition.

INTERIOR BULKHEADS: The interior teak bulkheads were finely fit where sighted.

WATER INTRUSION SIGNS: One minor area in the main salon showing water intrusion from long past (dry).

STORAGE AREAS: The cabinets, lockers, drawers, and shelving were well crafted and finely fit where sighted.

HEADLINERS: Headliner material in the cabins appeared to be a light colored vinyl panels. Good condition.

DOORWAYS: Solid wood cabin and head doors throughout vessel. Appeared serviceable.

FABRIC AND CUSHIONS: The general appearance of the beige leather cushions and fabrics are in good condition.

Settee

Head sink

FLOOR AND WINDOW COVERINGS: Marine plywood cabin sole with bilge access hatches, floor covering is carpet in Fair condition.

ACCOMMODATIONS: V-berth forward accommodates two, and a master cabin with a double berth. Serviceable.

HEADS: A single electric head in the lower cabin to port. The MSD (Marine Sanitation Device) is a type III. There is also a mirror, modern sink and Corian countertop, and separate enclosed shower. All serviceable.

CABIN APPOINTMENTS

INTERIOR DESCRIPTION:*(continued)*

FAUCET FIXTURES: The faucet fixtures and sinks were operable in the shower, head, and in the galley, corrosion free..

LIGHT FIXTURES: 12 volt cabin lights throughout the vessel were operable.

CABIN FURNISHINGS: Twin berth in the forward v-berth with hanging locker and drawers, the master cabin amidships has a full sized berth with night stand a hanging closet.

CABIN SOLE: Teak & holly throughout lower cabin in good condition, and flooring in the main salon is carpeted plywood in fair condition.

VENTILATION: Both sides of the main cabin have a large sliding window, and the large door leading to the cockpit provide excellent ventilation.

AIR CONDITIONING UNITS: Two (2), 16K BTU for the main salon & 10K BTU for the lower cabin, Dometic reverse cycle self contained units, installed aft in the engine room.

CABIN HEATING: AC's are reverse cycle for heating the cabins. Heat cycle not tested.

TELEVISIONS: 50" Samsung on main salon forward bulkhead. Not tested.

STEREO, ETC.: SONOS stereo in the main salon, did not test.

CONDITION AND DEFICIENCIES: The overall house keeping for this vessel was above average. It reflects the care of a conscientious crew, with good sea keeping skills.

GALLEY

LOCATION: Galley is on the starboard side of the lower cabin, with storage and built in microwave, refrigerator and stove.

Galley

SINKS: A single stainless steel rectangle shaped sink in the galley and rectangular porcelin sink in the head. Condition good.

REFRIGERATION: Built in refrigerator with freezer 120 Vac unit by Frigidaire, Model FFHT1425VV03, Sn: 1K22176115. Serviceable.

STOVE/OVEN: Princess; Three (3) burner electric stove with oven, 120Vac. Serviceable.

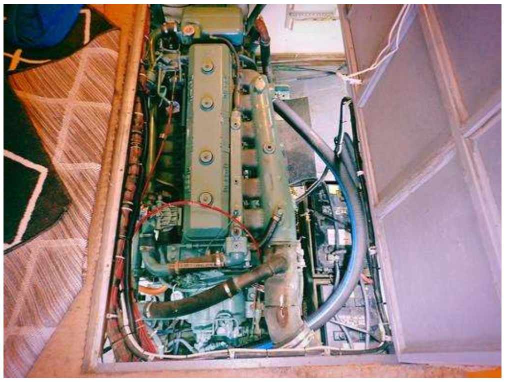
MICROWAVE: GE microwave oven built in to cabinet. Powers up.

COUNTERTOPS: Corian, in good condition.

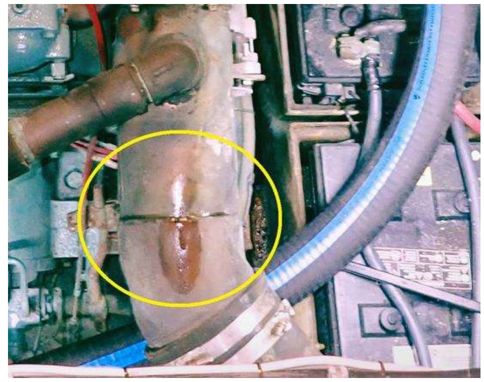
PROPULSION

MAIN ENGINES

TYPE: Diesel, Two (2) Inline-6 cyl., 2.99L , naturally aspirated engines.



Starboard engine



STARBOARD EXHAUST ELBOW LEAK

MANUFACTURER: Detroit Diesel, model 6-71M

SERIAL NUMBERS: Port Engine: Ser # 6A228915; Starboard Engine: Ser # 6A383263.

HORSE POWER: Each 280 HP @ 2300 RPM

INDICATED HOURS: Port hours 3300 , Starboard hours 3800.

THROTTLE CONTROLS: Morse mechanical lever/cable type, at flybridge helm station.

EMERGENCY SHUT DOWN: Port and starboard shut down stations for both air & Fuel at helm station clearly marked. Serviceable.

ENGINE MOUNTS AND BED: Main engine beds are FRP longitudinal stringers inboard and outboard. In conjunction, adjustable motor mounts are bolted to the stringers and are used to adjust the prop shaft alignment as well as secure the engines to the hull stringer structure. Serviceable with no signs of corrosion or deterioration.

DRIP PANS: None Sighted. Engine fluid and loose debris falls into the bilge area.

LUBRICATION: Level and Condition: Level indication is normal both port and starboard. Appears serviceable.

VENTILATION: Power blowers port and starboard. Natural, flow ventilation provided by hull vents. Appears adequate.

* **EXHAUST SYSTEM: [A1]** Raw water cooled 6" steel pipes and elbows to outboard frame supports. Flexible hose aft to FRP (fiber reinforced plastic) silencers located under the cockpit sole. Then exiting through fittings at transom. Hose to pipe connections are double clamped where sighted. Exhaust elbow on the starboard engine is leaking. (See Photo above)

INSULATION: Sound deadening insulation was noted in engine room. Appears serviceable.

PROP SHAFTS: Both stainless steel 1-3/4" diameter. Is serviceable.

ENGINE ALARMS: Low oil pressure alarm and coolant over heat warning audible at helm station. Appears serviceable.

ENGINE SHUT DOWN: Port and starboard shut down pull cables at helm station. Serviceable.

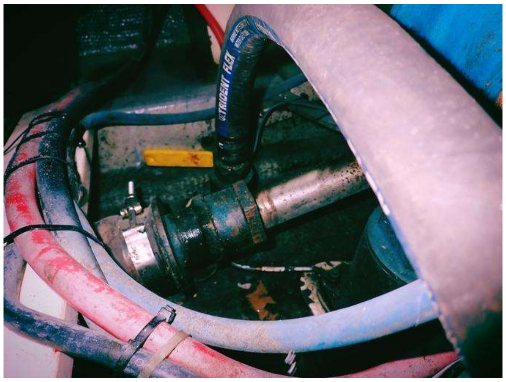
STUFFING BOX: Stuffing boxes and packing glands, are bronze single hex type, boots are well clamped and serviceable. Monitor Frequently for leakage and proper adjustment.

PROPULSION

MAIN ENGINES*(continued)*

* **STUFFING BOX:** ***(continued)***

Shaft seal stbd



Shaft seall port

CONDITION AND DEFICIENCIES: Engines and engine room clean, orderly, and well painted.

COOLING SYSTEM

TYPE: Freshwater reservoir type cooling with raw water cooled wet exhaust.

RAW WATER STRAINERS: All bronze alloy with sight glass for AC and generator, and each of the main engines. Serviceable.

COOLANT LEVEL: Normal level observed.

HOSES AND CLAMPS: Reinforced rubber hose clamped and well routed and supported, where sighted.

* **BELTS AND PULLEYS: [B1]** Belt condition on both engines appear serviceable. Port engine belt needs tightening.

SEACOCKS AND STRAINERS: Raw water seacocks were ball valve type. Serviceable

TRANSMISSIONS

TYPE: Twin Disc, 2:1 ratio, ID plates illegible. Serviceable.

FLUID LEVEL AND CONDITION: Normal level indicated on dipsticks. Appears serviceable.

CONTROLS: Morse type mechanical cable and linkage. Serviceable.

PROP SHAFT: Size: 1 3/4" Material: Stainless Steel.

FUEL SYSTEM

MAIN ENGINE(S) FUEL SYSTEM

FUEL TYPE: Diesel.

MATERIAL: Two (2) 200 gal. tanks, made of fiberglass. Serviceable with no evidence of leaks.

SECURED: Bracketed to stringers. Serviceable.

LOCATION: Fuel tanks are port and starboard, accessed from lazerette.

MANUFACTURING LABEL: None Sighted.

FUEL SYSTEM

MAIN ENGINE(S) FUEL SYSTEM*(continued)*

FILL PIPE LOCATIONS: Port gunwale for the port fuel tank, starboard gunwale for the starboard fuel tank. Both labeled "DIESEL". Serviceable.

FILL PIPE GROUNDED: Appears to be properly grounded.

FILL PIPE MATERIAL: Type B1 USCG approved hose. Appears serviceable.

HOSE CONNECTIONS, CLAMPS: Appears serviceable and approved where sighted.

FUEL LINES AND FITTINGS: Grade USCG type A1. Appears serviceable where sighted.

FUEL MANIFOLD VALVES: Ball type valves, operable.

VENT LOCATION: Topsides, flame screen sighted.

* **FUEL FILTERS: [A2]** Yes. Both remote mounted Racor filter/water 1100 series turbo separator type and engine mount Detroit Diesel spin on/off type. All three Racor primary filters are lacking the necessary heat shields for their plastic bowls.

FILTER/FUEL CONDITION: Appears serviceable.

ELECTRICAL SYSTEMS

ELECTRICAL SYSTEM (DC SYSTEM)

VOLTAGE: Lead acid battery powered 12 volt system.

* **BATTERIES: [A3]** Four (4), 8D wet cell batteries, two wired in parallel as a starting bank for each diesel engine. Also a Group 27 12V wet cell battery for generator starting. None of the batteries are well secured in their boxes, their positive terminals are not protected against accidental shorting, and a "wing nut" is used on the generator start battery.

Battery bank port side

Genset start battery

MAIN BATTERY SWITCHES: Two (2) main battery switches of the rotary type "Blue Sea Systems" mounted under the helm station on the fly bridge, and a rotary shut-off in the engine room. Appeared serviceable.

PANEL: Overcurrent Protection: Circuit breakers. Location: Starboard main salon cabinet. Access: Serviceable.

ELECTRICAL SYSTEMS

ELECTRICAL SYSTEM (DC SYSTEM)*(continued)*

*** PANEL:** ***(continued)***

Main electrical panel

AC electrical panel

BREAKERS/FUSES: 12V breakers for various well labeled circuits, plus two (2) spares, values of breakers unknown. Serviceable

TYPE CONNECTORS: Round Lugs: Captive type, where sighted. Condition: Appears serviceable.

ROUTING/SUPPORT: Well supported and secured where sighted.

CHARGING SYSTEM: Alternators on main diesel engines, and a Pro Mariner 30A battery charger on the aft engine room bulkhead. Serviceable.

OUTLETS: Sighted in the fly bridge helm. Did not test.

TERMINAL BLOCKS: Plastic double sided terminal block with captive lugs, well mounted and supported where sighted.

CIRCUIT LOAD MONITORS: Analog Voltmeter and analog Amp meter, located on the electrical panel. Serviceable.

ELECTRICAL SYSTEM (AC SYSTEM)

SHORE POWER INLET: Number: Set of two (2) Marinco 30 amp. Location: Cockpit starboard side. Serviceable.

SHORE POWER: Cord: Two (2) 30' long white Vinyl: Yes Adapter(s):"Y" adapter 50 amp 125/250 volt to (2)125/30 amp. Condition: Appears serviceable.

AC SOURCE SELECTOR SWITCH: Switch type: Manual rotary type. AC / Generator: Manual selector switch for shore or ship power. Location: Main AC panel, main salon. Appears serviceable.

MAIN BREAKER: Number: Two (2) in the main electrical panel starboard side of main salon forward. Serviceable.

BRANCH BREAKERS: Number: Fourteen (14) individually switched branch breakers. Location: Main AC panel, main salon. Well marked.

CIRCUIT LOAD MONITORS: Voltage & Wattage analog meters on AC main & AC control panels. Serviceable.

CONNECTIONS (TYPE): Captive lug type. Appears serviceable where sighted.

WIRE TYPE (SIZE AND RATING): Size and rating, where sighted, appears well routed and supported, serviceable for intended use.

ELECTRICAL SYSTEMS

ELECTRICAL SYSTEM (AC SYSTEM)*(continued)*

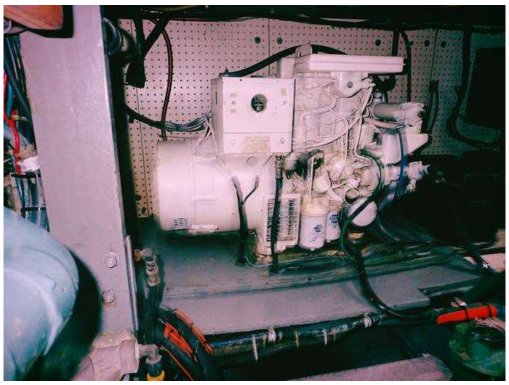
OUTLETS: Various AC outlets available throughout yacht, appear adequate and conveniently located. Tested ok for proper polarity. All GFCI (ground fault circuit interrupter) outlets and operational.

POLARITY: The polarity was checked by myself at all outlets that I could find and proved normal. Also Indicator light on each AC panel, appeared operational.

GALVANIC ISOLATOR: None Sighted. Highly recommended to reduce accelerated zinc loss.

GENERATORS AND INVERTERS

TYPE: Onan diesel generator, 120V/240V, Model MDJE, Ser. No. 06711308283. Serviceable.



Diesel generator

KILOWATT RATING: 7.5 kW, 60 cycle

VOLTAGE RATING: 120 /240 AC.

INDICATED HOURS: 4000. hours on meter.

LOCATION: Engine room, centerline aft.

FLUID LEVELS: Coolant normal. Oil normal.

COOLING SYSTEM: Freshwater and raw water wet exhaust type.

FUEL FILTER: Bulkhead mounted primary Racor 500 series, engine mounted secondary fuel filter, screw-on type. Serviceable.

EXHAUST SYSTEM: Aqua lift type FRP (fiber reinforced plastic). Raw water cooled, double clamped where sighted. Appears serviceable.

ACCESSIBILITY: Good.

WARNING LABELS: None Sighted.

FRESH WATER SYSTEM

FRESH WATER SYSTEM: (POTABLE WATER)

STORAGE TANKS: One (1) main ships tank, believed to be fiberglass, is located on centerline in the forward bilge.

CAPACITY: 150 gallons total

ACCESS: Access to tank appears adequate.

FRESH WATER SYSTEM

FRESH WATER SYSTEM: (POTABLE WATER)*(continued)*

INSPECTION/CLEANING ACCESS: Yes, serviceable.

FILL PIPE LOCATION: Starboard side weather deck forward, marked for water.

VENT PIPE LOCATION: Appears to be starboard topsides.

ACCUMULATOR TANK: None Sighted.

PUMPS: Jabsco diaphragm demand pressure switch pump, approx. 4.3 gpm, mounted in engine room forward. Serviceable.

FILTERS: None Sighted.

HOSES AND CLAMPS: Plastic tubing at various areas throughout vessel. Appears serviceable where sighted.

DOCK SIDE PRESSURE REGULATOR: None Sighted.

FRESH WATER SYSTEM (HOT WATER SYSTEM)

TYPE: 110 electric. Marine grade. Unknown Brand, approximately 11 gallon capacity. Serviceable.

PRESSURE RELIEF VALVE: Yes, copper pressure relief valve built into tank. Drainage: Reinforced plastic hose led to bilge. Did not test relief valve.

SANITATION

SANITATION (BLACK WATER)

MANUFACTURER: Raritan Electric flush. Serviceable.

NUMBER OF HEADS: One (1), port side lower cabin.

M.S.D TYPE USCG SYSTEM: Certification Type: MSD U.S.C.G. Type III. (Holding tanks)

RAW WATER SUPPLY AND CLAMPS: Appears serviceable where sighted.

DISCHARGE HOSES AND CLAMPS: Double clamped and serviceable where sighted.

PUMP-OUT LOCATION: Starboard side forward deck, fitting marked for waste.

HOLDING TANK: Fiberglass, reported to be 125 gal. Serviceable.

SANITATION (GREY WATER)

BASINS, SHOWERS, HOSES AND CLAMPS: The basins on the vessel drain to topside thru-hulls in immediate area of basin. The shower drains to a sump in the forward bilge and is pumped overboard. Did not test.

PUMPS: Sump & Rule 500 GPH pump with float switch. Operable.

STEERING SYSTEM

STEERING SYSTEM

* **TYPE: [C1]** Hydraulic, by Hynautic, where sighted appeared serviceable. Number of Stations: one (1), on flybridge. Lines and fittings: Copper and reinforced flexible hose, with metallic fittings. Appears serviceable (no leaks or wet areas observed). Mounting: Cylinder and ram actuator are well secured under lazerette. Gauge on the steering hydraulic reservoir is reading 23 psi, recommended pressure is 30 psi.

ACTUATOR CYLINDER: Appears serviceable.

STEERING SYSTEM

STEERING SYSTEM *(continued)*

* **ACTUATOR CYLINDER:** ***(continued)***


Steering cylinder


Steering stbd

MOUNTING: Cylinder and ram actuator are well secured.

RUDDER STOCK: Stainless steel, 1-1/2" diameter.

RUDDER POSITION INDICATOR: VDO electro-mechanical type at starboard rudder post. Appears serviceable.

UPPER RUDDER BEARING SUPPORT: Both well mounted on a plywood base attached to stringers. Appears serviceable.

PACKING GLAND: Bronze hex nut type packing glands. Appeared serviceable. Monitor frequently.

GROUND TACKLE

GROUND TACKLE

* **ANCHORS: [A4]** A single 35 lb. Delta type mounted at the bow, attached to approximately 200 ft. of 5/16" galvanized BBB chain. Anchor to chain shackle pins are not seized.

WINDLASS: Simpson Lawrence mounted at bow. Serviceable.

SPARE ANCHOR(S): Two spare anchors stowed in a locker below the cockpit deck, a Fortress and a Danforth, both approximately 20 lbs. Serviceable. A 3/4" 3-strand nylon anchor rode is ready at the bow with approximately 150' stowed in a divided anchor locker for emergency deployment. Serviceable.

NOTE: I recommend that the anchor and chain be removed from the chain locker, closely examined for condition, exact length, and attachment to the vessel.

ELECTRONICS AND NAVIGATION EQUIPMENT

ELECTRONICS AND NAVIGATION EQUIPMENT

VHF: Standard Horizon Explorer. Powered up.

RADAR: New Furuno Navgen VX2 25 Kw 6' open array system, in the box to be installed.

CHART PLOTTER: GARMIN GPSmap 2010c chartplotter, powers up.

ELECTRONICS AND NAVIGATION EQUIPMENT

ELECTRONICS AND NAVIGATION EQUIPMENT*(continued)*

* **CHART PLOTTER:** ***(continued)***

Nav instruments

Bridge radios

AUTOHELM: Furuno NavPilot at helm station. Serviceable.

* **COMPASSES: [B2]** One (1) 5" Danforth at the bridge, no deviation cards present. Helm compass is low on oil.

ANTENNAS: All antennas sighted appear to be well mounted and serviceable.

OTHER: Bennett trim tab electric hydraulic system. Not tested.

ELECTRONICS (ENTERTAINMENT)

STEREO SYSTEM: A Sonos stereo receiver with speakers. Did not test.

TELEVISION(S): Samsung 50" smart TV. Did not test.

THRU-HULLS

THRU-HULLS:

NOTE: All thru hulls sighted had minimal corrosion and operated freely. Most were recently replaced.

NOTE: Because this was an in-water survey, one or more thru hulls may have been missed during inspection.

BONDED: They were bonded where sighted. Appears serviceable.

BONDING SYSTEM

BONDING SYSTEM

MAIN BONDING CONDUCTOR: The bonding system is mostly well established where sighted. A separate bonding system was not performed and I did not use a corrosion meter to establish the level of protection. However the bonding system is using individual green insulated wire and appeared to be serviceable were sighted. I also noted bolts inside for a transom mounted zinc amidships. Monitor it frequently for condition and adequate protection.

SAFETY EQUIPMENT

SAFETY EQUIPMENT (UNITED STATES COAST GUARD)

NUMBER AND TYPE OF PFD'S: Sixteen (16) Type II-U.S.C.G. approved, two are the inflatable type.

NUMBER OF THROWABLE PFD'S: Single Type IV-U.S.C.G. approved throwable device. Life ring on bridge.

SAFETY EQUIPMENT

SAFETY EQUIPMENT (UNITED STATES COAST GUARD)*(continued)*

* **FIRE EXTINGUISHERS: [A5]** Two (2) Type B-II (5lbs) and one (1) Type B-I dry chemical with gauges. All fire extinguishers are stowed properly and visible. The B-1 fire extinguisher is under a factory recall.

* **VISUAL DISTRESS SIGNALS: [A6]** Day/night visual distress signals are hand held flares. Expiration date: 2016

SOUND DEVICES: Yes, air horn and whistle. Operable.

NAVIGATION LIGHTS: Sidelights are operable.
Sternlight is operable.
Anchor light is ioperable.

* **INLAND NAVIGATION RULE BOOK (12M-39'4" OR LONGER): [A7]** Not sighted.

"NO OIL DISCHARGE" PLAQUE: Yes, found properly displayed.

TRASH DISPOSAL PLACARD: Yes, found properly displayed in main salon area.

* **WASTE MANAGEMENT PLAN (OVER 40'): [A8]** None Sighted.

AUXILIARY SAFETY EQUIPMENT

LIFE RAFT: Not sighted.

E.P.I.R.B.: None Sighted. But highly recommended.

* **FIXED FIRE EXTINGUISHING SYSTEM (HALON TYPE): [C2]** Two Fireboy Halon-type systems, one forward and one aft in the engine space. No Inspection tags present.

SEARCH LIGHT: Remote Jabsco searchlight operated from the helm station. Serviceable.

* **FUME SNIFFER ALARM SYSTEMS: [A9]** Three (3) CO detectors/smoke detectors installed but believed to be outdated.

BILGE PUMPS

LIST: Four (4) Rule 1500 gph bilge pumps with float switches and manual switches, located in the forward, mid, and aft bilges. Serviceable.

AIR CONDITIONING AND HEAT (AIR CONDITIONING)

AIR CONDITIONING AND HEAT (AIR CONDITIONING)

TYPE: Two (2) unitized reverse cycle type by Cruisair. Both are located along the aft engine room bulkhead, starboard side. One is a model: RX16 115/60/1, Sn: 74064151, 16K BTU, for the main salon, the second is a model: RX10 115/60/1, Sn: 73971076, 10K BTU, for the lower forward cabin. Both serviceable.

THRU-HULL STRAINER: Yes, Groco sight glass style. In the engine room starboard side. Appears serviceable.

HOSES, CLAMPS AND CONNECTORS: Appear to be adequately sized and serviceable where sighted.

RAW WATER COOLING PUMP: 110 volt MARCH/Dometic electric pump, system is equipped with a seacock and sea strainer assembly. Serviceable.

DRIP TRAYS: Yes, one for each unit. Appear serviceable.

IV. FINDINGS AND RECOMMENDATIONS

Deficiencies noted under **"SAFETY"** should be addressed before vessel is next underway. These findings represent an endangerment to personnel and/or the vessel's safe and proper operating condition. ***Findings may also be in violation of U.S.C.G. regulations.*** Deficiencies noted under **"OTHER DEFICIENCIES"** should be corrected in the near future so as to maintain standards and to help the vessel to retain it's value.

Deficiencies will be listed under the appropriate heading:

A. SAFETY DEFICIENCIES
B. OTHER DEFICIENCIES NEEDING ATTENTION
C. SURVEYORS NOTES AND OBSERVATIONS

A. SAFETY DEFICIENCIES:

FINDINGS	*RECOMMENDATIONS*

A.1 (PAGE 10) EXHAUST SYSTEM:

Exhaust elbow on the starboard engine is leaking.	*Investigate further and repair or renew as necessary.*

A.2 (PAGE 12) FUEL FILTERS:

All three Racor primary filters are lacking the necessary heat shields for their plastic bowls.	*Install heat shields to comply with USCG Safety Regulation 33 CFR 183.590, and ABYC H33.5.6: Fire Test "....must withstand a flame for 2 1/2 minutes."*

A.3 (PAGE 12) BATTERIES:

None of the batteries are well secured in their boxes, their positive terminals are not protected against accidental shorting, and a "wing nut" is used on the generator start battery.	*Replace wing nut with a hex nut, secure batteries and install protective covers over positive terminals. CFR 183.420, CFR 183.445, NFPA 302 7-43.*

A.4 (PAGE 16) ANCHORS:

Anchor to chain shackle pins are not seized.	*Apply siezing wire to shackle pins to prevent loosening.*

A.5 (PAGE 18) FIRE EXTINGUISHERS:

The B-1 fire extinguisher is under a factory recall.	*Return for new fire extinguisher(s). Consumer Contact: Kidde toll-free at 855-271-0773 or online at www.kidde.com and click on Product Safety Recall for more information.*

A.6 (PAGE 18) VISUAL DISTRESS SIGNALS:

Day/night visual distress signals and hand held flares. Out of date.	*Comply with USCG regulations for Visual Distress Signals: 33 CFR 175.101*

A.7 (PAGE 18) INLAND NAVIGATION RULE BOOK (12M-39'4" OR LONGER):

Inland Navigation Rule Book not on board.	*Comply with USCG Safety Regulations concerning Navigation Rules: "The operator of a vessel 39.4 feet or greater is responsible for having and maintaining a copy of the Navigation Rules on board while operating on U.S. inland waters".*

IV. FINDINGS AND RECOMMENDATIONS

A. SAFETY DEFICIENCIES:

A.8 (PAGE 18) WASTE MANAGEMENT PLAN (OVER 40'):

FINDINGS	*RECOMMENDATIONS*
Waste Management Plan not on board.	*Comply with USCG Safety Regulations for polution: 33 CFR 155. "....vessels 40 ft or longer equipped with a galley and berthing must have a written waste management plan describing procedures for collecting, processing, storing, and discharging garbage, and must designate the person in charge of carrying out the plan."*

A.9 (PAGE 18) FUME SNIFFER ALARM SYSTEMS:

FINDINGS	*RECOMMENDATIONS*
Three (3) CO detectors/smoke detectors installed but believed to be outdated.	*Comply with ABYC recommendations for smoke and fume detection: 24.6.1 Carbon monoxide detectors shall be installed on all boats with an enclosed accommodation compartment(s). One in each living and sleeping space. Renew after 5 years.*

B. OTHER DEFICIENCIES NEEDING ATTENTION:

B.1 (PAGE 11) BELTS AND PULLEYS:

FINDINGS	*RECOMMENDATIONS*
Port engine belt needs tightening.	*Investigate further and tighten or renew as necessary.*

B.2 (PAGE 17) COMPASSES:

FINDINGS	*RECOMMENDATIONS*
Helm compass is low on oil.	*Investigate further and repair or renew as necessary.*

C. SURVEYOR'S NOTES AND OBSERVATIONS:

C.1 (PAGE 15) TYPE:

FINDINGS	*RECOMMENDATIONS*
Gauge on the steering hydraulic reservoir is reading 23 psi, recommended pressure is 30 psi.	*Investigate further and pump up air pressure as necessary and monitor.*

C.2 (PAGE 18) FIXED FIRE EXTINGUISHING SYSTEM (HALON TYPE):

FINDINGS	*RECOMMENDATIONS*
No Inspection tags present on the FireBoy fixed fire extinguishing systems.	*Maintain and inspect per manufacturer's instructions, usually inspect every two years with initials on an inspection tag.*

NOTE: If cruising more than 25 nautical miles offshore it is also recommended that a USCG approved self-inflating life raft be fitted to the vessel, and a first aid kit and small manual watermaker be added to the ship's safety gear.

V. SUMMARY AND VALUATION

STATEMENT OF OVERALL VESSEL RATING OF CONDITION:

It is the surveyor's experience that develops an opinion of the **OVERALL VESSEL RATING OF CONDITION** After a the survey has been completed and the findings have been organized in a logical manner.

The grading of condition, developed by **BUC RESEARCH**, and accepted in the marine industry, for a vessel at the time of survey, determines the adjustment to the range of base values in the **BUC USED BOAT PRICE GUIDE**, for a similar vessel sold within a given time period, as a consideration to determine the Market Value.

The following is the accepted marine grading system of condition:
"EXCELLENT (BRISTOL) CONDITION", is a vessel that is maintained in mint or bristol fashion - usually better than factory new - loaded with extras - a rarity.
"ABOVE AVERAGE CONDITION", has had above average care and is equipped with extra electrical and electronic gear.
"AVERAGE CONDITION", ready for sale requiring no additional work and normally equipped for her size.
"FAIR CONDITION", requires usual maintenance to prepare for sale.
"POOR CONDITION", substantial yard work required and devoid of extras.
"RESTORABLE CONDITION", enough of hull and engine exists to restore the boat to usable condition.

As a result of my investigation, as shown in the **SYSTEMS AND FINDINGS AND RECOMMENDATIONS** section of this **REPORT OF SURVEY,** and by virtue of my experience, my opinion is

OVERALL VESSEL RATING: ABOVE AVERAGE

STATEMENT OF VALUATION:

1. The **"FAIR MARKET VALUE"** is the most probable price in terms of money which a vessel should bring in a competitive and open market under all conditions requisite to a fair sale, the buyer and seller, each acting prudently, knowledgeably and assuming the price is not affected by undue stimulus.

Implicit in this definition is the consummation of a sale as of a specified date and the passing of title from seller to buyer under conditions whereby:
a. Buyer and seller are typically motivated.
b. Both parties are well informed or well advised, and each acting in what they consider their own best interest.
c. A reasonable time is allowed for exposure in the open market.
d. Payment is made in terms of cash in U.S. dollars or in terms of financial arrangements comparable thereto; and
e. The price represents a normal consideration for the vessel sold unaffected by special or creative financing or sales concessions granted by anyone associated with the sale.

The "Sales Comparison Approach to Value" is the method used to determine Fair Market Value.: By averaging the asking price of similar Hateras & Bertram sport fishing boats, located in the Eastern half of the US, the average price adjusted for age (using the Marshal Scale) is $89,000. Selling prices for a vessel of this type and age vary a great deal depending on upgrades, care, and condition. The BUC book value of this boat in "above average" condition and with location considerations, is between $78,600. and $86,300.

Therefore, after consideration of the reliability of the data, the extent of the necessary adjustments and condition of the vessel, it is your surveyor's opinion that the **"FAIR MARKET VALUE"** of the subject vessel is:

$86,600 Dollars
Eighty Six Thousand Six Hundred Dollars

2. The **"ESTIMATED REPLACEMENT COST"** indicates the retail cost of a new vessel of the same make/model with similar equipment offered by the same manufacturer. **"ESTIMATED REPLACEMENT COST"** of the subject vessel is:

$2,025,000 Dollars
Two Million Twenty Five Thousand Dollars

V. SUMMARY AND VALUATION

SUMMARY:

In accordance with the request for a marine survey of the vessel "LONGSHANKS", for the purpose of evaluating its present condition and estimating its Fair Market Value and Replacement Cost, I herewith submit my conclusion based on the preceding report. "LONGSHANKS" was personally inspected by the undersigned on August 20, 2024, and was found to be a well appointed, well maintained, and well constructed vessel. Subject to correction of deficiencies listed in section IV A. (Safety), the vessel is considered to be suitable for its intended use of coastal cruising and fishing. Other deficiencies listed should be attended to as soon as possible to maintain value and functionality.

SURVEYOR'S CERTIFICATION:

I certify that, to the best of my knowledge and belief:

The statements of fact contained in this report are true and correct.

The reported analyses, opinions, and conclusions are limited only by the reported assumptions and limiting conditions, and are my personal, unbiased professional analyses, opinions, and conclusions.

I have no present or prospective interest in the vessel that is the subject of this report, and I have no personal interest or bias with respect to the parties involved.

My compensation is not contingent upon the reporting of a predetermined value or direction in value or direction in value that favors the cause of the client, the amount of the value estimate, the attainment of a stipulate result, or the occurrence of a subsequent event.

I have made a personal inspection of the vessel that is the subject of this report.

This report is submitted without prejudice and for the benefit of whom it may concern.

ATTENDING SURVEYOR:

Ronald E. Varg

SOCIETY OF ACCREDITED MARINE SURVEYORS®
Ronald Varg
AMS® # 1324

Ronald E. Varg, Surveyor SAMS, AMS

VI. PHOTOGRAPHS

View stbd fwd

Foredeck

View stbd side

View of stern

VI. PHOTOGRAPHS

Anchor platform

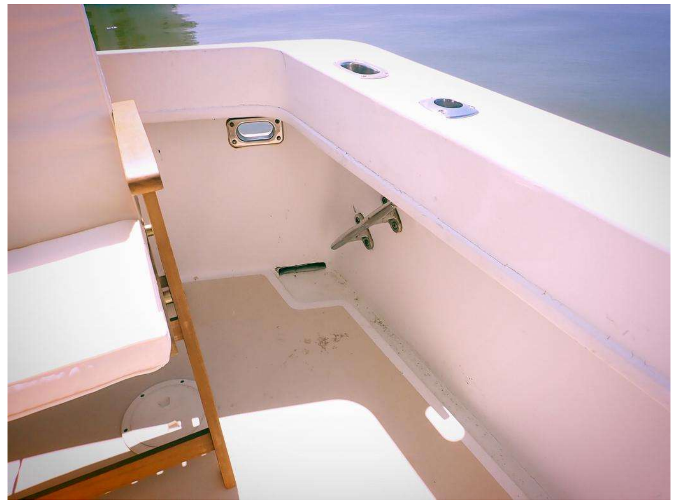
Fuel filter

Lower cabin access

Helm area

VI. PHOTOGRAPHS

Electrical inlet



Cockpit port side

Boat placard & battery switch

Hull No. 319 (barely visible)